SlugRoute

Title Page

Project Title: SlugRoute
Date: 04-07-2026
Sponsored by: Self-sponsored / Baskin Engineering
Team Name: [Insert Team Name]

Team Members:

- Juan (Product Owner)
- Jon (Initial Scrum Master)
- Alexis, Finn

SlugRoute

Problem / Opportunity
UCSC students often have lectures in one building but labs and sections in completely different, obscure locations. Navigating these transitions—especially early in the quarter—is stressful.

Who cares?
The 19,000+ UCSC student body, especially students with complex STEM schedules.

Who does it help?
Students trying to find hidden lab rooms or forest-path shortcuts between back-to-back classes.

Why?
Standard GPS apps treat a course as a single destination.
SlugRoute treats a course as a **collection of meetings across campus.**

Project Scope

Goals

- **Comprehensive Data Scraping:** Extract lectures, labs, and sections from PISA
- **Multi-Meeting Visualization:** Show 0-N pins per course
- **Visual Wayfinding:** Color-coded pins + building and meeting info
- **Intelligent Routing:** Real-time pathfinding from user GPS and between pins

---

Sprint 1

User Stories

- As a user, I want to access a fast-loading website so that I can view course information instantly without waiting for long server responses. 5
- As a student, I want to see a comprehensive and up-to-date list of class times and locations so that I can plan my schedule without checking multiple websites. 13
- As a user, I want the course data to be stored accurately and available every time I return to the site. 1
- As a user, I want a stable, well-maintained app built using professional standards so that I receive regular updates, fewer bugs, and a reliable experience. 2

Spikes

- Go learning curve (specifically Gin)
- Proper way to store our scraped data

Infrastructure

- Initialize GitHub
- Set up Go (Gin)
- Install SQLite3 and setup schemas
- Build Python scraper prototype

Sprint 2

User Stories

- As a user, I want a fast and responsive interface so that I can interact with my course data in real-time without the entire page having to reload. 5
- As a student, I want to enter a course and see color-coded pins on a building. 8
- As a student, I want to add all my classes to one view so I can visualize my entire weekly commute across campus. 3
- As a student, I would like to be able to add/delete courses from the current map. 1

Spikes

- Normalize UCSC building names to coordinates
- SQL queries
- Google Maps Markers API

Infrastructure

- Deploy SQLite3 and create necessary tables
- Manually collect latitude and longitude data to minimize using the Geocoding API

Sprint 3

User Stories

- As a student, I want real-time GPS tracking on the map. 2
- As a student, I want to click a pin and see meeting information. 5
- As a student, I want a walking path to my next meeting. 3
- As a student, I want to track up to five courses simultaneously so that I can focus on planning a standard full-time academic semester without cluttering my view. 2

Spikes

- Google Maps Routing API
- UI layout

Infrastructure

- Potentially store image assets
- Integrate Google Maps Routing API

Sprint 4

User Stories

- As a student, I want paths between back-to-back classes. 2
- As a mobile user, I want a responsive, scalable UI. 13
- As a student, I would like to be able to save classes to my local profile. 3
- As a user, I want to be able to request directions from my current location or between pins. 8

Spikes

- Google Maps Directions API
- UI/UX polish
- Cross-browser testing

Infrastructure

- Leverage IndexedDB and/or LocalStorage API for user data persistence
- Integrate Google Maps Directions API

Architecture

Data Tier
Python (BeautifulSoup) scraper → SQLite3

Logic Tier
Go (Golang) API for relational queries

Presentation Tier
HTML, CSS, JavaScript + Google Maps API for rendering

Data Model
1-to-Many: Course → Multiple meetings

Technologies

- **Backend:** Go (Golang)
  - Gin framework

- **Frontend:** HTML + JavaScript + CSS
  - Vanilla languages
- **Scraping:** Python
  - Requests, BeautifulSoup
- **Database:** SQLite3
  - Serverless, lightweight
- **Maps:** Google Maps API
  - Comprehensive, Routing, Directions
- **DevOps:** GitHub Actions
  - Source code management

Challenges / Risks

- TBD lab/section locations → need placeholder handling
- Needing to find the latitude and longitude of classrooms
- Using non-copyrighted building photos
- Some of the API we are planning to implement has limited free use
- Learning multiple APIs (Google Maps, etc.)
- Overlapping pins → clustering or offset logic
- Go learning curve
- UCSC terrain → unofficial paths not in standard APIs
- Expanding the UI without cluttering the user environment

MVP

Description
A web app where users search a course and see all meeting locations with:

- Colored pins
- Building and detailed meeting information
- Walking routes from current GPS

User Stories

1. Search for a class and see all meeting pins
2. Pins are color coordinated
3. User location is shown on the map
4. Clicking a pin shows meeting info and potentially a photo

Product Backlog

1. Allows users to submit their own pictures and have an approval system for it
2. Add user accounts to not rely local storage/cookies
3. Add longitude and latitude
4. Add a photo for the entrances of classrooms if applicable
5. Use Google Maps API to take our application customizable
6. Add notifications for when meetings start (toggle on/off)
7. Add floor plans to search for specific rooms.
8. Allow the user to choose Paths depending on time, incline, or type of path.
9. Implement partial search handling so they can look for differing lectures based on number or description (Ex. CSE, 30, MAT)(Sanitizer)